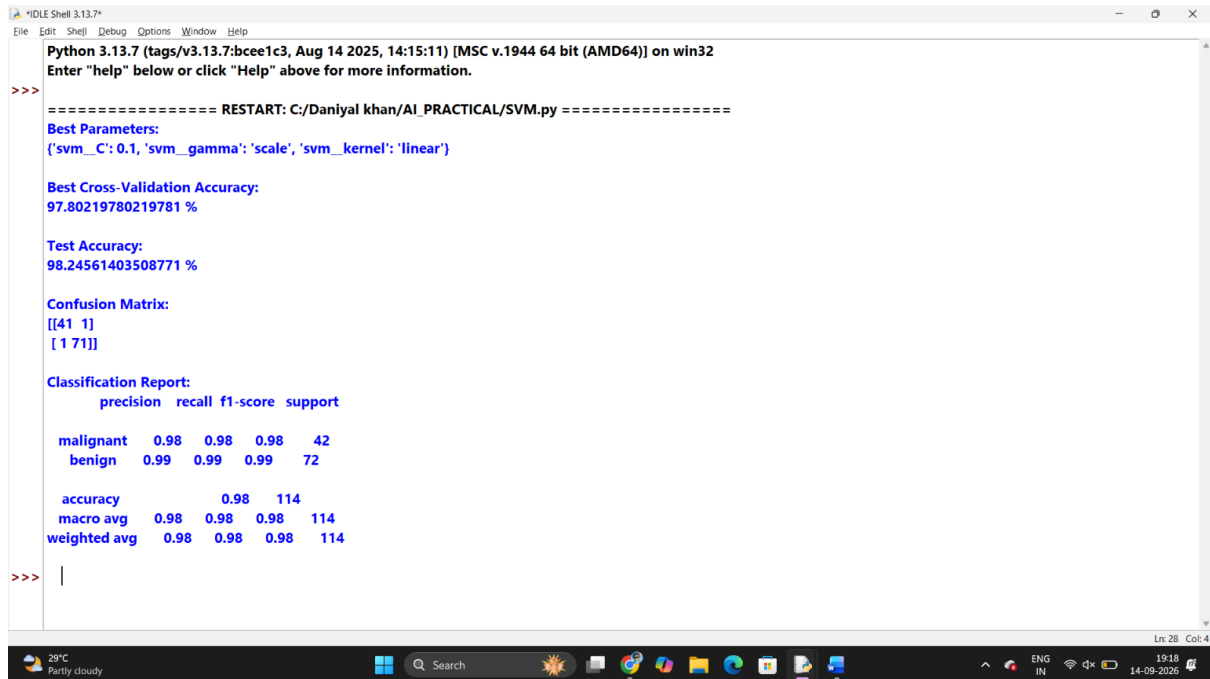


SHETH LUJ COLLEGE

Support Vector Machines (SVM)

- ❑ Implement the SVM algorithm for binary classification.
- ❑ Train an SVM model using a given dataset and optimize its parameters.
- ❑ Evaluate the performance of the SVM model on test data and analyze the results.



The screenshot shows a Python IDLE Shell window with the following output:

```
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Daniyal khan/AI_PRACTICAL/SVM.py =====
Best Parameters:
{'svm_C': 0.1, 'svm_gamma': 'scale', 'svm_kernel': 'linear'}

Best Cross-Validation Accuracy:
97.80219780219781 %

Test Accuracy:
98.24561403508771 %

Confusion Matrix:
[[41  1]
 [ 1 71]]

Classification Report:
      precision    recall  f1-score   support

malignant    0.98    0.98    0.98    42
benign       0.99    0.99    0.99    72

accuracy          0.98    114
macro avg    0.98    0.98    0.98    114
weighted avg    0.98    0.98    0.98    114

>>> |
```

The window title is "IDLE Shell 3.13.7". The taskbar at the bottom shows the date as 14-09-2026 and time as 19:18.